

3 (a) State the first law of thermodynamics.

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..... [1]

(b) An adiabatic process is one in which no heat is supplied to or extracted from a system.

- (i)** Determine the change in the internal energy of an ideal gas when the gas expands and does 500 J of work in an adiabatic process.

change in internal energy = J [1]

- (ii)** Hence, describe how the temperature of the gas in **(i)** will change at the end of the adiabatic process.

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..... [1]

(c) 2.5 mol of an ideal gas in another system is heated up from 300 K to 500 K without any change in volume.

The molar heat capacity of the gas is numerically equal to the quantity of thermal energy required to raise the temperature of 1.0 mol of the gas by 1.0 K.

- (i)** Determine the molar heat capacity of the gas.

molar heat capacity = J mol⁻¹ K⁻¹ [3]

- (ii)** Explain why the molar heat capacity would be larger if the heating takes place at constant pressure rather than at constant volume.

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..... [2]

4 Fig. 4.1 shows an electron with a horizontal velocity of $5.0 \times 10^7 \text{ m s}^{-1}$ entering a region